

A Machine Learning Approach for Early Detection of Postpartum Depression in Bangladesh

Jasiya Fairiz Raisa¹[0000–0002–9548–8154], M. Shamim Kaiser²[0000–0003–4721–8798], and Mufti Mahmud^{3,4,5}[0000–0002–2037–8348]

¹ Information and Communication Technology, Bangladesh University of Professionals, Dhaka 1216, Bangladesh
jasiyafairiz@gmail.com

² Institute of Information Technology, Jahangirnagar University, Savar, Dhaka 1342, Bangladesh
mskaiser@juniv.edu

³ Department of Computer Science, Nottingham Trent University, Clifton Campus, Nottingham NG11 8NS, UK

⁴ Medical Technologies Innovation Facility, Nottingham Trent University, Clifton Campus, Nottingham NG11 8NS, UK

⁵ Computing and Informatics Research Centre, Nottingham Trent University, Clifton Campus, Nottingham NG11 8NS, UK
mufti.mahmud@ntu.ac.uk, muftimahmud@gmail.com

Abstract. Postpartum depression is a severe mental health issue exhibited among perinatal women after the childbirth process. While the negative impact of postpartum depression is extensive in developing countries, there is a significant lack of proper tools and techniques to predict the disorder due to negligence. This work proposes a machine learning-based system for finding the risk factors and prevalence of postpartum depression in Bangladesh. We developed a survey of different socio-demographic questions and modified questions from two standard postpartum depression screening scales (EPDS, PHQ-2). Data from 150 women have been collected, processed, and implemented in different machine learning models to find—the best performing models. Based on the collected data of the perinatal women in Bangladesh, the best performing machine learning model was Random Forest. The performance metrics for the best model were AUC: 98%, Accuracy: 89%, and Sensitivity: 89%. The performance of the models varies from 88%-98% (AUC), 82%-89% (Accuracy), and 81%-89% (Sensitivity). We have also found the top risk factors for causing PPD. According to this work, the prevalence of PPD in Bangladesh is 66.7% (Considering the medium and high chance of PPD). This proposed work is the first to detect the risk factors and prevalence of PPD in Bangladesh using a machine learning approach.

Keywords: Depression · Postpartum Depression · Machine Learning · Detection model · Mental health.

1 Introduction

Postpartum depression (PPD) is a non-psychotic mental disorder that is commonly seen in mothers within the first year of delivery [22],[12]. PPD often goes undiagnosed in most cases, and it can lead to severe complications and should be managed promptly [1]. Women with PPD may experience significant difficulties in cognitive and emotional processes that impact mother-infant attachment. Globally, the prevalence of postpartum depression among perinatal women varies between 0.5 and 60.8% [12]. The prevalence of PPD in Bangladesh is 39.4% in the first twelve months after childbirth [4]. Identifying women at risk of having PPD is essential for clinical practice because it enables targeted prevention treatments [23]. A robust PPD classifier can assist health care practitioners in identifying and effectively managing at-risk individuals [19], as evidenced by recent studies examining this prospect and feasibility [17]

Machine learning (ML) assists in making precise estimates using data from numerous sources and has been utilized in recent years in prediction studies [20]. ML is thought to assist mental health practitioners in more objectively defining mental diseases than the Diagnostic and Statistical Manual of Mental Disorders [25] and identifying these illnesses to improve the efficacy of therapies [11].

Considering the effectiveness of ML in predictive studies and the unique case of PPD, ML-based classification can be applied for prevalence and risk factors detection for PPD, providing benefit to the mothers and their families.

Given the severity of the problem of PPD, the number of existing works for detecting it is not significant. Furthermore, most of the existing works are from the perspective of developed countries, which might not be effective from the perspective of developing countries such as Bangladesh. To our knowledge, no such work currently exists that detects PPD in Bangladesh using machine learning approaches.

The work aims to develop a machine learning-based system for risk analysis and detection of postpartum depression among Bangladeshi perinatal women and a survey questionnaire prior to data collection.

The contribution of the study can be summarized as follows:

1. Development and comparison of different machine learning models for detecting PPD among Bangladeshi women
2. Finding the correlation between demographic factors and postpartum depression in Bangladesh

2 Related works

The number of existing works to detect postpartum depression among women is not significantly rich. In the perspective of Bangladesh, some research works presented the prevalence and risk factors from a sociological perspective. While narrowing down the existing works in the machine learning perspective, the number of works drops low in numbers accordingly. The existing works in this

field are primarily from the perspective of developed countries [23, 21, 27, 2]. A significant number of the existing works are developed using data collected from the hospital or any cohort studies. In this section, we have reviewed existing works for PPD detection, first from the Bangladesh perspective. Later, some significant works on machine learning-based PPD detection are reviewed.

2.1 PPD in Bangladesh Perspective

Postpartum depression is one of the understudied research topics in Bangladesh. Several studies have analyzed the prevalence and risk factors among Bangladeshi perinatal women.

The study developed by Gausia et al. [9] has reported the prevalence and associated factors of postpartum depression in the rural areas of Bangladesh. The authors found the prevalence of PPD at 22% and the incidence of postpartum depressive symptoms at 9.8% at 6-8 weeks postpartum.

Using a cohort study approach, Edhborg et al. [15] studied the incidence and risk factors of PPD in Bangladesh. The researchers analyzed the mental and socio-economic conditions of 588 women during pregnancy and after childbirth. After implementing multiple Cox's regression, it was found that 18.58% of the participants were at risk of postpartum depression. The socio-economic status, history of abuse and anxiety during pregnancy, and previous depressive symptoms are the most critical risk factors for postnatal depression.

Azad et al. [4] developed a cross-sectional study to find the prevalence and related risk factors in the slum areas of Dhaka city. The research shows that the prevalence of PPD among the participants is 39.4% within the first 12 months after childbirth. The authors used EPDS as the standard scale for measuring the risk of PPD. The risk factors with the highest impact were found to be the job status.

A study [18] developed using hospital data found that 65.22% of the participants were at risk of PPD. They found that the mother's age, multigravida, newborn's gender, and congenital anomalies are the most critical factors responsible for the prevalence of PPD.

2.2 Machine Learning for PPD Detection

Researchers are now approaching to predict and classify many health problems with machine learning, including postpartum depression is one. Some notable works are reviewed to analyze their implemented techniques and PPD classification accuracy.

Zhang et al. [26] proposed that SVM and FFS-RF were found to have the best prediction effects for PPD. They performed a longitudinal survey to collect data from 508 women and used EPDS for measuring the risk of PPD.

Zhang et al. [27] implemented EHR datasets for detecting PPD among perinatal women. They found that logistic regression with L2 regularization was the best performing algorithm using data up to childbirth. For the data after childbirth, MLP performed the best.

The PRAMS 2012-2013 dataset and the PHQ-2 questionnaire were proposed to be implemented in several machine learning algorithms by Shin et al. [21] to find the prevalence of PPD. The authors found Random Forest to be the best-performing algorithm for predicting postpartum depression.

Andersson et al. [2] developed several machine learning models and used data collected from hospitals in Sweden. The extremely Randomized Trees model was the best performing model for this large population-based study.

Another research work was developed using data collected from hospitals by Tortajada et al. [23]. The proposed work used MLP for predicting PPD and achieved 81% accuracy. De Choudhury [8] performed a longitudinal online survey on postpartum women and experimented with different regression models on the dataset.

A comparison among Functional-gradient boosting, Decision-trees, Naive Bayes and SVM for PPD detection is shown by Nataranjan et al. [16]. The authors developed a longitudinal dataset for this research work and showed that Functional-gradient boosting performed best among the implemented algorithms. Wang et al. [24] developed a machine learning-based model on EHR data and found that SVM performed best with their dataset.

3 Materials and Methodology

The development of the PPD screening survey and the methodology of the proposed work are discussed in this section. The whole workflow of the model is depicted in figure 1.

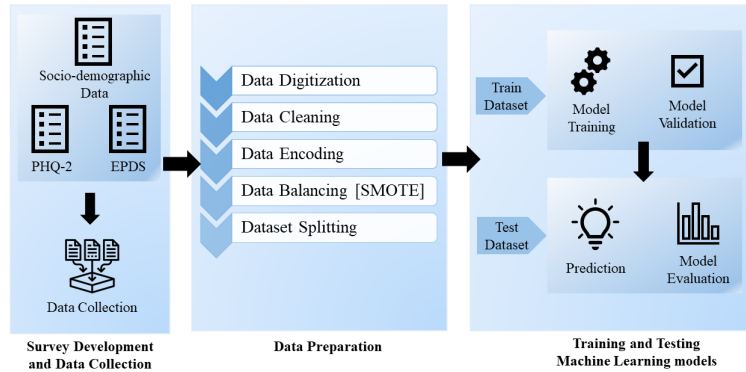


Fig. 1. Proposed workflow of the PPD detection model

3.1 Sampling

Study Population All reproductive age group women who gave birth in the last 24 months in Bangladesh.

Study Location The data collection was performed in different regions of Bangladesh. We included women from separate residences, education, and income level.

Study Type A cross-sectional study was performed by interviewing 150 postpartum women. The complete questionnaire is developed in Bangla for the ease of the participants.

Time Period The data was collected from January 2022 to February 2022 in different geographical locations in Bangladesh.

Inclusion and Exclusion Criteria All mothers who gave birth before the interview and their latest born child is alive and not more than two years old were included in the study. The study only included postpartum women who gave their consent to this interview. Those mothers who are seriously ill and unable to respond during data collection were excluded from this study.

3.2 Measures

The interview questionnaire contained socio-demographic questions, a PHQ-2 questionnaire, and the EPDS questionnaire.

Socio-demographic Questionnaire Questions about general health, relationships, education, income level, and history of abuse are developed and added to the questionnaire.

PHQ-2 The Patient Health Questionnaire-2 [14] is used to determine the mothers' mental health condition before, during, and after their latest childbirth using only two questions for each time period.

EPDS The Edinburgh Postnatal Depression Scale [7] is a ten-item self-report assessment, that is used for in-depth screening for PPD among the participants. This self-screening survey has been verified as a clinical screening tool for postpartum depression in Bangladeshi women aged 6–8 weeks [7, 10]. Several psychologists validated the questionnaire before starting the data collection process.

3.3 Data Collection Procedure

A pilot study was performed on 8 participants (5% of the 150 participants), and the data from the pilot study were excluded from the original dataset. The data was collected in person from different areas of Bangladesh, including different cities and villages. However, in some cases, we had to collect data from appropriate participants online since it was hard to reach postpartum women during the lockdown. The collected data were cross-checked to ensure the completeness of the questionnaire by the participants.

3.4 Data Preparation

At first, the collected data was transferred in digital format. Since the data were cross-checked for completeness, no missing-value entries were found in the raw dataset. We translated the data from Bangla to English to fit the model. The data were encoded into numeric values. The PHQ2 questionnaire for the different time periods of mothers' lives was labelled "likely depressive" or "not likely depressive" based on their results and the cutoff score. Similarly, the data of the EPDS questionnaire were scored and added to calculate their sum for getting the final EPDS score. In this study, we have considered three levels of likely to have postpartum depression:

1. Score 0 to 8: Less likely to have PPD (Marked as "Low")
2. Score 9-12: Likely to have PPD (Marked as "Medium")
3. Score 13-30: Highly likely to have PPD (Marked as "High")

3.5 Statistical Approach for Data Analysis

The processed data were analyzed, grouped by the level of EPDS score range, using the statistical analysis tool, SPSS. Independent sample T-test assuming unequal variances and Chi-Square test were used for numerical and categorical variables as appropriate.

3.6 Machine Learning Approach for PPD Detection

Resampling to Address Dataset Imbalance Unbalanced data can reduce the accuracy of ML-based classification [6]. To correct the data imbalance, we employed the widely used synthetic minority oversampling approach (SMOTE) [5]. Instead of reproducing existing minority members, SMOTE develops synthetic members based on Euclidean distances between data points in feature space.

Model Development We have implemented several supervised machine learning algorithms and ensemble models for PPD detection in the proposed work. After reviewing the existing works, we have selected some of the most common and best-performing models for the detection of postpartum depression. We

have used SVM, Random Forest, and Logistic Regression from the supervised techniques and boosting classifiers as the ensemble methods, such as Gradient Boosting and Extreme Gradient Boosting for PPD detection in this work. The algorithms were selected in such a way that over-fitting or under-fitting problems don't affect the result. Both classical and ensemble algorithms were implemented to ensure the validity of the performance of the model using the collected dataset.

Model Training for Classification The processed dataset is split into an 80-20 ratio as training and testing datasets. Five ML algorithms were individually implemented to detect PPD in this proposed work. A 10-fold cross-validation strategy was used to evaluate the classification models. The original samples were randomly partitioned into ten equal-sized subsamples, and a single subsample was retained as validation data for testing the model built using the other nine subsamples.

Model Performance Evaluation The test dataset was fed into the trained machine learning models for prediction. The original and the predicted data were compared to develop a confusion matrix. A series of performance metrics were obtained for each model, including Area under Curve, Sensitivity, Accuracy, Precision, F1 Score, and Geometric Mean. Among these metrics, Area Under Curve (AUC), accuracy, and sensitivity are the three most important metrics for PPD detection. These are used to evaluate the effects of each model and choose the best prediction model.

Table 1. Risk Factors for Postpartum Depression

Risk Factor	χ^2 (df)	p-value	Risk Factor	χ^2 (df)	p-value
Age Range	6.77(6)	0.342	Loss During Pregnancy	15.6(14)	0.337
Residence	0.931(2)	63%	Abuse	15.0(2)	< .001
Education Level	4.61(6)	60%	Sharing Feeling With Friends	5.26(2)	0.072
Marital Status	NaN	NaN	Number Of Latest Pregnancy	6.96(8)	0.541
Job Before Pregnancy	7.74(12)	0.805	Pregnancy Length	8.72(10)	0.558
Income Before Pregnancy	5.83(8)	67%	Pregnancy Planned	1.33(2)	0.515
Current Job	11.2(12)	0.509	Checkup	1.63(2)	0.443
Current Income	4.80(8)	0.779	Fear Of Pregnancy	6.04(2)	0.049
Husband's Education	8.03(6)	0.236	Diseases During Pregnancy	9.08(4)	6%
Husband's Income	6.02(8)	0.645	Age Of Latest Child	3.94(6)	0.685
Addiction	2.01(2)	0.365	Age Of Previous Baby	23.3(18)	18%
Number Of Children	4.57(6)	0.6	Delivery Mode	1.64(2)	44%
Disease Before Pregnancy	3.05(4)	0.549	Birth Complicay	2.43(2)	0.296
Pregnancy Loss	2.81(4)	0.59	Gender	0.539(2)	0.764
Family Type	0.354(2)	0.838	Breastfeed	4.69(2)	0.096
Number Of Family Member	12.7(20)	0.89	Illness Of The Baby	14.5(2)	< .001
Relation With In-laws	16.1(4)	0%	Worry For Baby	5.62(2)	0.06
Relation With Husband	20.3(4)	<.001	Rest While Baby Is Monitored	2.25(2)	0.325
Relation With Baby	4.00(4)	0.406	Rest When Baby Sleeps	12.4(2)	0.002
Husband And Baby Relation	9.06(4)	6%	Anger After Delivery	44.0(2)	< .001
Motherhood Feeling	11.1(4)	3%	Work Feeling After Delivery	13.3(6)	0.039
Received Support	25.0(16)	0.07	PHQ 2(Before Pregnancy)	1.13(2)	0.567
Want Support	33.7(8)	< .001	PHQ 2(During Pregnancy)	6.68(2)	0.035

4 Result

4.1 Correlation of the Socio-demographic Features with PPD

The status of postpartum depression significantly differed by many factors [3, 13]. We have collected 46 socio-demographic features of postpartum women for this research work. These factors were statistically analyzed to understand better these features and their effect on the occurrence of PPD. The distribution of predictor variables are summarized in Table 1. From the analysis, it is found that factors like relation with husband, need for support, history of abuse, illness of the newborn, depression prior to pregnancy, etc., influence the occurrence of PPD among women.

4.2 Prevalence of PPD

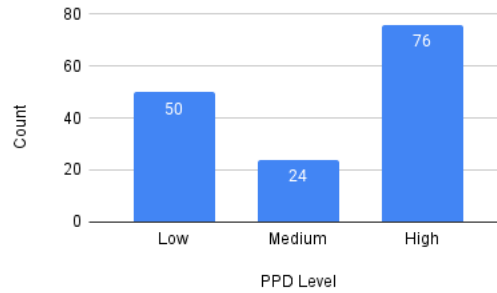


Fig. 2. Prevalence of PPD in Bangladesh

Using the trained model to predict PPD in the test dataset yielded the highest AUC-ROC of 98% and the highest accuracy of 89% with the Random Forest model. Among the 150 participants, 100 (66.7% of the total) were regarded as likely to have PPD, considering the EPDS score over the cutoff score (medium and high risk). The data distribution is shown in figure 2 15.33% of the participants were at risk of depression prior to their latest pregnancy, and 25.33% were at risk of depression during pregnancy. Participants with previous depressive symptoms had a higher chance of PPD (73.91% more in the participants with previous depression before pregnancy and 76.32% more in the participants with previous depression during pregnancy) than women with no history of depression.

4.3 Effect of Machine Learning Classifier Models

Classification models were developed to predict the multiclass of postpartum depression (High, Medium, Low). The classification process was performed in

the balanced dataset developed using SMOTE technology. The PPD prediction models were established using five ML algorithms Random Forest, Extreme Gradient Boosting, SVM, Gradient Boosting, and Logistic Regression. The algorithms were applied to the training data set, and later test data set was fed into the model to predict the PPD level. The performance metrics for the implemented models are shown in the table 2, and the ROC-AUC curve and the confusion metrics of the best performing model, Random Forest, are illustrated in the figure 3.

Table 2. Performance metrics of the Top Five Machine Learning Models

Classifier	AUC	Acc	Sens	Prec	F1 Score	G-Mean	Cross Val
Random Forest	98%	89%	89%	89%	89%	89%	63%
Extreme Gradient Boosting	94%	86%	85%	86%	85%	85%	55%
SVM	88%	84%	84%	86%	84%	84%	63%
Gradient Boosting	91%	84%	81%	83%	82%	81%	66%
Logistic Regression	91%	82%	83%	82%	82%	83%	57%

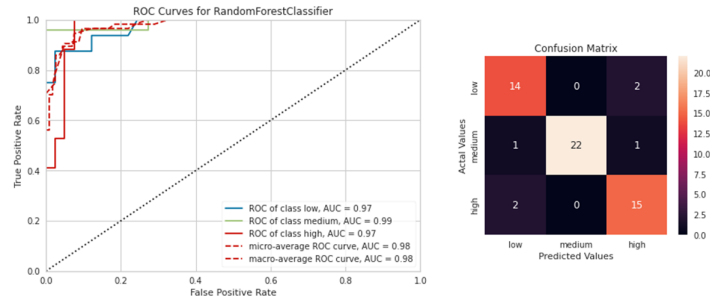


Fig. 3. Performance metrics of the Random Forest model

5 Discussion

5.1 Key Findings and Contributions

In the current work, we implemented supervised and ensemble techniques with the data of postpartum women from Bangladesh. All the implemented algorithms achieved similar performance, among which the Random Forest and the Extreme gradient boosting algorithms had the best performance. The overall performance of the machine learning algorithms ensures that the data properly fit into the model.

The prevalence of postpartum depression among the studied women is found to be about 66.67%, considering the medium and high probability of PPD based

on the EPDS scale. The random forest algorithm calculated the prevalence of PPD as 70.17%.

From the provided statistical analysis of the dataset, the correlation between the demographic factors and PPD can be measured. The features can be categorized based on feature importance calculated from their p-value.

With significantly high-performance metrics, (Accuracy: 89%, AUC: 98%, Sensitivity: 89%), the proposed model can be said to be significant and promising in predicting postpartum depression. To our best knowledge, this is the first machine learning-based model to predict postpartum depression among Bangladeshi perinatal women.

5.2 Limitations

It is important to compare the proposed work with the existing relevant works for performance validation of the prior work. However, because earlier studies employed various types of data and offered various machine learning algorithms to detect the prevalence of PPD, the model prediction impacts cannot simply be compared. Additionally, potential selection bias, small sample size, and the requirement of more extended validation may have affected the performance of the models.

6 Conclusion

This study aimed to compare a variety of machine learning algorithms for predicting perinatal women at risk of postpartum depression. We sought to identify potential risk factors for developing postpartum depression, while early detection of PPD. To understand this study's quality, it is necessary to compare the results and contributions of the proposed work. In the literature review section, we have seen the lack of existing works based on cross-sectional studies and the implementation of several standardized scales for PPD screening. Drawing a comparison among the existing works with this study, it can be said that this research work has addressed these gaps in the existing works accordingly. Despite the novelty of the current work, there are some limitations pointed out in the discussion section. Further studies should develop to address these gaps. It is evident that machine learning can considerably improve the early diagnosis of PPD. The proposed work's findings imply a potential benefit of utilizing machine learning to screen new mothers for PPD in the perspective of Bangladesh. This work's data-driven, machine learning-based strategy can be expanded to create an accurate and scalable system for cost-effectively offering preventive treatments, assuring improved mental healthcare and support for postpartum mothers in Bangladesh.

References

1. Aliani, R., Khuwaja, B.: Epidemiology of Postpartum Depression in Pakistan: A Review of Literature. *National Journal of Health Sciences* **2**(1), 24–

- 30 (Feb 2017). <https://doi.org/10.21089/njhs.21.0024>, http://njhs.21089.com/wp-content/uploads/2017/02/6-Razia-Alini_M_S.pdf
2. Andersson, S., Bathula, D.R., Iliadis, S.I., Walter, M., Skalkidou, A.: Predicting women with depressive symptoms postpartum with machine learning methods. *Scientific Reports* **11**(1), 7877 (Dec 2021). <https://doi.org/10.1038/s41598-021-86368-y>, <http://www.nature.com/articles/s41598-021-86368-y>
 3. Ay, F.: Postpartum depression and the factors affecting It: 2000-2017 Study results. *Journal of Psychiatric Nursing* (2018). <https://doi.org/10.14744/phd.2018.31549>, https://www.journalagent.com/phd/pdfs/PHD-31549-RESEARCH_ARTICLE_AY.pdf
 4. Azad, R., Fahmi, R., Shrestha, S., Joshi, H., Hasan, M., Khan, A.N.S., Chowdhury, M.A.K., Arifeen, S.E., Billah, S.M.: Prevalence and risk factors of postpartum depression within one year after birth in urban slums of Dhaka, Bangladesh. *PLOS ONE* **14**(5), e0215735 (May 2019). <https://doi.org/10.1371/journal.pone.0215735>, <https://dx.plos.org/10.1371/journal.pone.0215735>
 5. Chawla, N.V., Bowyer, K.W., Hall, L.O., Kegelmeyer, W.P.: SMOTE: Synthetic Minority Over-sampling Technique. *Journal of Artificial Intelligence Research* **16**, 321–357 (Jun 2002). <https://doi.org/10.1613/jair.953>, <http://arxiv.org/abs/1106.1813>, arXiv: 1106.1813
 6. Chawla, N.V.: DATA MINING FOR IMBALANCED DATASETS: AN OVERVIEW. *DATA MINING AND KNOWLEDGE DISCOVERY HANDBOOK* p. 15
 7. Cox, J.L., Holden, J.M., Sagovsky, R.: Detection of postnatal depression. Development of the 10-item Edinburgh Postnatal Depression Scale. *The British Journal of Psychiatry: The Journal of Mental Science* **150**, 782–786 (Jun 1987). <https://doi.org/10.1192/bjp.150.6.782>
 8. De Choudhury, M., Counts, S., Horvitz, E.J., Hoff, A.: Characterizing and predicting postpartum depression from shared facebook data. In: *Proceedings of the 17th ACM Conference on Computer Supported Cooperative Work and Social Computing*. p. 626–638. CSCW '14, Association for Computing Machinery, New York, NY, USA (2014). <https://doi.org/10.1145/2531602.2531675>, <https://doi.org/10.1145/2531602.2531675>
 9. Gausia, K., Fisher, C., Ali, M., Oosthuizen, J.: Magnitude and contributory factors of postnatal depression: a community-based cohort study from a rural subdistrict of Bangladesh. *Psychological Medicine* **39**(6), 999–1007 (Jun 2009). <https://doi.org/10.1017/S0033291708004455>
 10. Gausia, K., Fisher, C., Algin, S., Oosthuizen, J.: Validation of the Bangla version of the Edinburgh Postnatal Depression Scale for a Bangladeshi sample. *ECU Publications* **25** (Nov 2007). <https://doi.org/10.1080/02646830701644896>
 11. Graham, S., Depp, C., Lee, E.E., Nebeker, C., Tu, X., Kim, H.C., Jeste, D.V.: Artificial Intelligence for Mental Health and Mental Illnesses: an Overview. *Current Psychiatry Reports* **21**(11), 116 (Nov 2019). <https://doi.org/10.1007/s11920-019-1094-0>
 12. Halbreich, U., Karkun, S.: Cross-cultural and social diversity of prevalence of postpartum depression and depressive symptoms. *Journal of Affective Disorders* **91**(2-3), 97–111 (Apr 2006). <https://doi.org/10.1016/j.jad.2005.12.051>
 13. Insan, N., Weke, A., Forrest, S., Rankin, J.: Social determinants of antenatal depression and anxiety among women in South Asia: A systematic review & meta-analysis. *PLOS ONE* **17**, e0263760 (Feb 2022). <https://doi.org/10.1371/journal.pone.0263760>
 14. Kroenke, K., Spitzer, R.L., Williams, J.B.W.: The Patient Health Questionnaire-2: validity of a two-item depression screener. *Medical Care* **41**(11), 1284–1292 (Nov 2003). <https://doi.org/10.1097/01.MLR.0000093487.78664.3C>

15. Maigun Edhborg, H.E.N.: Incidence and Risk Factor of Postpartum Depressive Symptoms in Women: A Population Based Prospective Cohort Study in a Rural District in Bangladesh. *Journal of Depression and Anxiety* **04**(02) (2015). <https://doi.org/10.4172/2167-1044.1000180>, <http://www.omicsgroup.org/journals/incidence-and-risk-factor-of-postpartum-depressive-symptoms-in-women-a-population-based-prospective-cohort-study-in-a-rural-district-in-bangladesh-2167-1044-1000180.php?aid=51561>
16. Natarajan, S., Prabhakar, A., Ramanan, N., Bagilone, A., Siek, K., Connelly, K.: Boosting for Postpartum Depression Prediction. In: 2017 IEEE/ACM International Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE). pp. 232–240. IEEE, Philadelphia, PA, USA (Jul 2017). <https://doi.org/10.1109/CHASE.2017.82>, <http://ieeexplore.ieee.org/document/8010637/>
17. Righetti-Veltima, M., Conne-Perréard, E., Bousquet, A., Manzano, J.: Risk factors and predictive signs of postpartum depression. *Journal of Affective Disorders* **49**(3), 167–180 (Jun 1998). [https://doi.org/10.1016/s0165-0327\(97\)00110-9](https://doi.org/10.1016/s0165-0327(97)00110-9)
18. Rn, D.R.M., Rn, M.: Depression Among Postnatal Mothers in Bangladesh p. 8
19. Robertson, E., Grace, S., Wallington, T., Stewart, D.E.: Antenatal risk factors for postpartum depression: a synthesis of recent literature. *General Hospital Psychiatry* **26**(4), 289–295 (Aug 2004). <https://doi.org/10.1016/j.genhosppsych.2004.02.006>
20. Shatte, A.B.R., Hutchinson, D.M., Teague, S.J.: Machine learning in mental health: a scoping review of methods and applications. *Psychological Medicine* **49**(9), 1426–1448 (Jul 2019). <https://doi.org/10.1017/S0033291719000151>
21. Shin, D., Lee, K.J., Adeluwa, T., Hur, J.: Machine Learning-Based Predictive Modeling of Postpartum Depression. *Journal of Clinical Medicine* **9**(9), 2899 (Sep 2020). <https://doi.org/10.3390/jcm9092899>, <https://www.mdpi.com/2077-0383/9/9/2899>
22. Stewart, D.E., Robertson, E., Phil, M., Dennis, C.L., Grace, S.L., Wallington, T.: POSTPARTUM DEPRESSION: LITERATURE REVIEW OF RISK FACTORS AND INTERVENTIONS p. 289
23. Tortajada, S., García-Gómez, J.M., Vicente, J., Sanjuán, J., de Frutos, R., Martín-Santos, R., García-Esteve, L., Gornemann, I., Gutiérrez-Zotes, A., Canellas, F., Carracedo, , Gratacos, M., Guillamat, R., Baca-García, E., Robles, M.: Prediction of Postpartum Depression Using Multilayer Perceptrons and Pruning. *Methods of Information in Medicine* **48**(03), 291–298 (2009). <https://doi.org/10.3414/ME0562>, <http://www.thieme-connect.de/DOI/DOI?10.3414/ME0562>
24. Wang, S., Pathak, J., Zhang, Y.: Using Electronic Health Records and Machine Learning to Predict Postpartum Depression. *Studies in Health Technology and Informatics* **264**, 888–892 (Aug 2019). <https://doi.org/10.3233/SHTI190351>
25. Wisner, K.L., Moses-Kolko, E.L., Sit, D.K.Y.: Postpartum depression: a disorder in search of a definition. *Archives of women’s mental health* **13**(1), 37–40 (Feb 2010). <https://doi.org/10.1007/s00737-009-0119-9>, <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4426488/>
26. Zhang, W., Liu, H., Silenzio, V.M.B., Qiu, P., Gong, W.: Machine Learning Models for the Prediction of Postpartum Depression: Application and Comparison Based on a Cohort Study. *JMIR Medical Informatics* **8**(4), e15516 (Apr 2020). <https://doi.org/10.2196/15516>, <http://medinform.jmir.org/2020/4/e15516/>
27. Zhang, Y., Wang, S., Hermann, A., Joly, R., Pathak, J.: Development and validation of a machine learning algorithm for predicting the risk of postpartum depression among pregnant women. *Journal of Affective Disorders* **279**, 1–8 (Jan 2021). <https://doi.org/10.1016/j.jad.2020.09.113>, <https://linkinghub.elsevier.com/retrieve/pii/S0165032720328093>